Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Planetary Systems

Mara Attia

Kapteyn Astronomical Institute
University of Groningen

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Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above T_{eff}
- Evaluate atmospheric escape mechanisms and retention criteria

Recap: Lecture 4

- Metal–silicate separation driven by magma oceans
- Hf–W chronometry: Earth's core formed in ~30–60 Myr
- Goldschmidt classification: siderophile, lithophile, chalcophile, atmophile
- Dynamo requirements: conducting fluid + convection + $\mathrm{Rm} \gtrsim 10$–100
- Mars lost its dynamo between ~4.1 and 3.7 Ga → atmospheric erosion

Today: The thin gaseous envelope that controls a planet's surface conditions.

1. Atmospheric composition

Earth’s atmosphere, ISS (2009). Credit: NASA

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- Requires mass $\gtrsim$ 5–10 $M_\oplus$ to capture nebular gas
- Disk disperses in ~3–10 Myr (Lecture 2)
- **Gas giants:** Jupiter and Saturn, massive $\mathrm{H_2}$/He envelopes
- **Ice giants:** Uranus and Neptune, envelopes only 10–20% of total mass
- Terrestrial planets too small → any primordial H quickly lost

Secondary atmospheres

Produced by outgassing from the interior.

- Volcanism and magma ocean degassing release dissolved volatiles
- Speciation depends on **oxygen fugacity** (melt's O_2 redox state; Lecture 4):
  - Oxidising: CO_2, H_2O, N_2
  - Reducing: H_2, CO, N_2
- **Venus:** CO_2-dominated (96.5%), 92 bar
- **Mars:** CO_2-dominated (95%), but only 6 mbar
- **Titan:** thick N_2 (from NH_3 conversion), 1.5 bar

Where do atmospheres come from?

A pictorial summary of the three classes.

Primary	Directly captured nebular gas; H_2/He dominated. Requires $M \gtrsim 5\text{–}10\, M_\oplus$ before disk dispersal.
Secondary	Outgassed from the interior during accretion + cooling. Composition set by f_{O_2} of the mantle.
Tertiary	Modified by surface processes, photochemistry, and (for Earth) biology.

Over time, a planet's atmospheric inventory can be **replaced** or **re-equilibrated** entirely: the present composition may not reflect the initial inventory.

Tertiary atmospheres

Modified by surface processes, photochemistry, or biology.

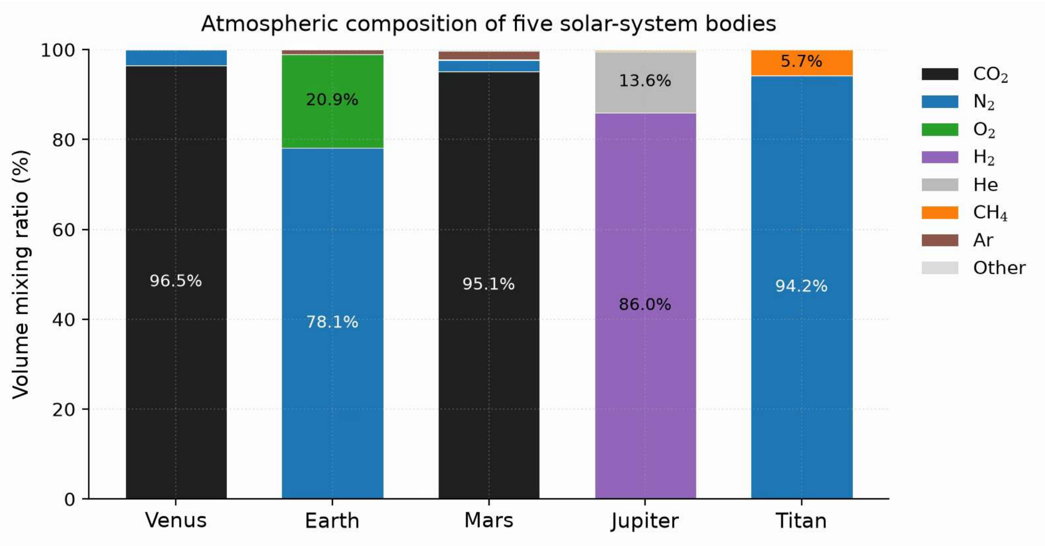
- Earth's original outgassed atmosphere: likely $\mathrm{CO_2}$ + $\mathrm{N_2}$ (similar to Venus)
- Oxygenic photosynthesis transformed it over billions of years
- Present atmosphere: $\mathrm{N_2}$ (78%) + $\mathrm{O_2}$ (21%)
- $\mathrm{O_2}$ **is entirely biogenic**, and would disappear in ~Myr without life

Earth's atmosphere is a **tertiary atmosphere**: the product of billions of years of biological modification.

Comparative atmospheric properties

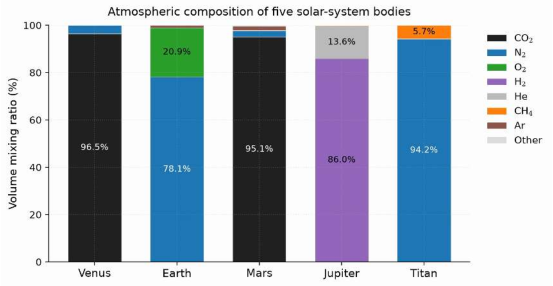
	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	CO_2	N_2	CO_2	H_2	N_2
Secondary gas	N_2	O_2	N_2	He	CH_4
μ	43.4	29.0	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 737 K.



Atmospheric composition by volume for five solar-system bodies. Course figure.

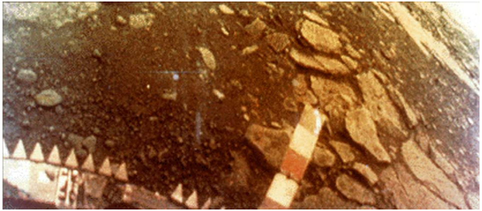
Reading the composition bars



- Venus and Mars: $\mathrm{CO_2}$-dominated secondary atmospheres, despite pressures differing by four orders of magnitude.
- Earth's $\mathrm{N_2}$/$\mathrm{O_2}$ mix is biogenically modified (tertiary); Jupiter keeps its primary $\mathrm{H_2}$/He envelope.
- Titan's massive $\mathrm{N_2}$ atmosphere (with 5% $\mathrm{CH_4}$) is unusual for such a small body.

Three origins, five outcomes: an atmosphere's composition encodes its history.

Venus: a secondary atmosphere in extremis



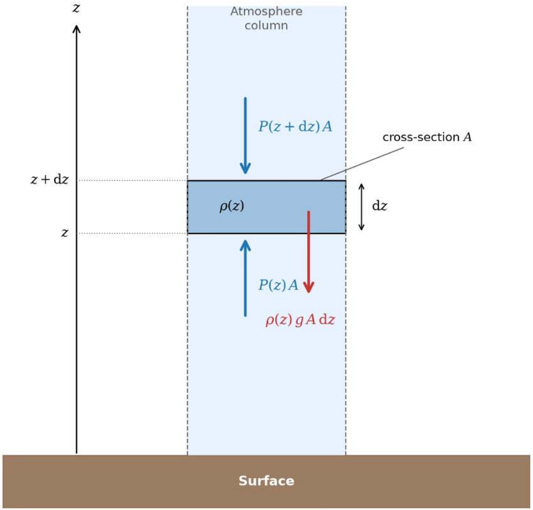
Credit: USSR/NASA NSSDC, public domain

- *Venera 13* lander, 1 March 1982
- Surface pressure: 92 bar
- Surface temperature: 737 K
- Orange sky: thick $\mathrm{CO_2}$ atmosphere scatters light
- Lander survived 127 minutes

A vivid demonstration of how a massive secondary atmosphere transforms surface conditions.

2. Hydrostatic equilibrium

Earth’s atmosphere. Credit: NASA/ISS



Force balance on a thin slab of area A and thickness $\mathrm{d}z$: pressure on the lower face pushes up, pressure on the upper face and the slab's own weight push down. Course figure.

Hydrostatic equilibrium

Consider a thin slab of atmosphere at height z, thickness $\mathrm{d}z$, area A:

- Pressure from below (upward): $P(z) \cdot A$
- Pressure from above (downward): $P(z + \mathrm{d}z) \cdot A$
- Weight (downward): $\rho(z)\, g\, A\, \mathrm{d}z$

Force balance:

$$P(z)\, A - P(z + \mathrm{d}z)\, A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Pressure decreases with altitude: each layer supports the weight above it.

The ideal gas law (atmospheric form)

To solve hydrostatic equilibrium, we need $\rho(P)$:

$$P = n\,k_\mathrm{B}T = \frac{\rho\,k_\mathrm{B}T}{\mu\,m_u}$$

n	number density
k_B	Boltzmann constant (1.381×10^{-23} J K^{-1})
μ	mean molecular weight (dimensionless, in amu)
m_u	atomic mass unit (1.661×10^{-27} kg)

Rearranging: $\rho = P\,\mu\,m_u/(k_\mathrm{B}T)$

The barometric formula

Substitute $\rho = P\mu m_u/(k_\mathrm{B}T)$ into hydrostatic equilibrium:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B}T}\, P$$

For an **isothermal** atmosphere (constant T, constant g), this separates:

$$\boxed{P(z) = P_0\, \exp\!\left(-\frac{z}{H}\right)}$$

where $H = k_\mathrm{B}T/(\mu\, m_u\, g)$ is the **pressure scale height**.

- Pressure drops by $e \approx 2.718$ every scale height
- 99% of the atmospheric mass lies below $4.6\,H$ for an isothermal column

3. Blackboard derivation: scale height

Earth’s atmosphere. Credit: NASA/ISS

Deriving the scale height

Goal: Derive $H = k_\mathrm{B}T/(\mu\, m_u\, g)$ from hydrostatic equilibrium + ideal gas law, and compute H for solar system bodies.

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_\mathrm{B} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\, \mu\, m_u}{k_\mathrm{B} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B} T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}}$$

Physical interpretation:

- **Higher** T → larger H: hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H: harder to loft
- **Stronger gravity** g → smaller H: more compressed

Key insight: H encodes the competition between thermal energy and gravitational binding.

Worked example: Earth's scale height

$$H_\oplus = \frac{k_\mathrm{B} T}{\mu\, m_u\, g} = \frac{1.381 \times 10^{-23} \times 288}{28.97 \times 1.661 \times 10^{-27} \times 9.81}$$

$$= \frac{3.98 \times 10^{-21}}{4.72 \times 10^{-25}} \approx 8400 \text{ m} \approx 8.4 \text{ km}$$

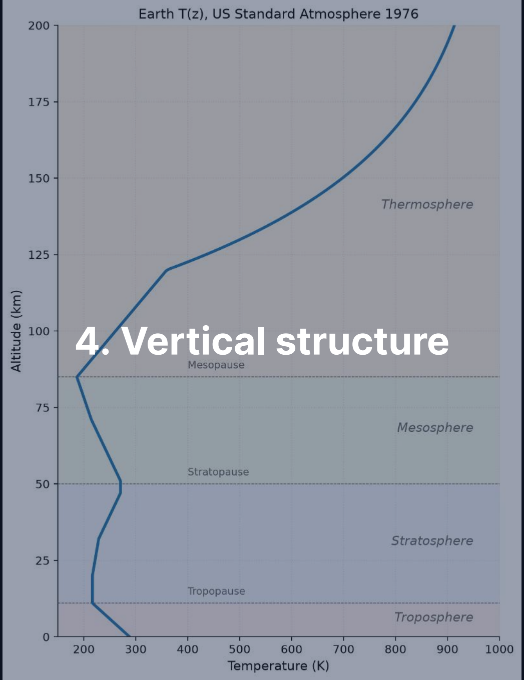
- Commercial aircraft: 10–12 km (~1.2–1.4 H), pressure ~0.2–0.3 atm
- Top of Everest: 8.8 km (~1 H), pressure ~0.3 atm
- 99% of mass below 4.6 H; colder layers aloft compress this to ~30 km

Scale heights across the solar system

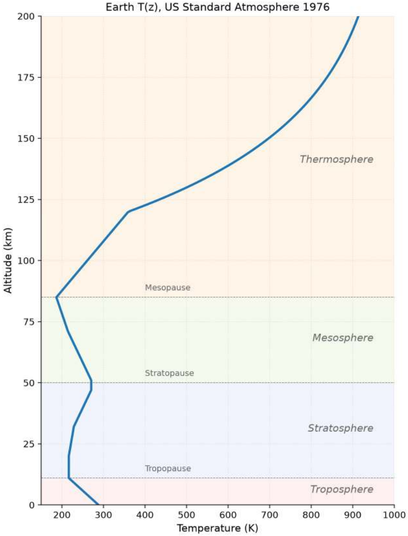
Body	T (K)	μ	g (m s^{-2})	H (km)
Venus	737	43.4	8.87	15.9
Earth	288	29.0	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

- Jupiter: large H despite strong gravity, because H_2 is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot but heavy molecules + decent gravity → moderate H

4. Vertical structure

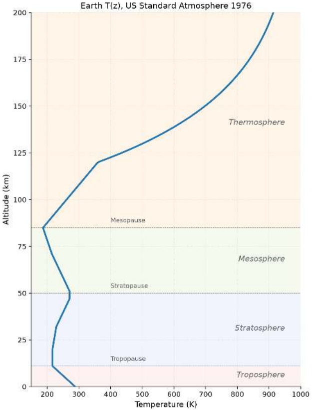


Earth’s atmospheric layers. Course figure



Earth's vertical temperature profile with the four named layers, US Standard Atmosphere 1976. Course figure.

Atmospheric layers



Credit: Course figure; US Standard Atmosphere 1976

Layers defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$:

1. **Troposphere** (0–12 km): T decreases with z
   Heated from below; convective
2. **Stratosphere** (12–50 km): T increases
   UV absorbed by ozone layer
3. **Mesosphere** (50–85 km): T decreases
   Coldest point at mesopause (~190 K)
4. **Thermosphere** (>85 km): T increases
   EUV (extreme-ultraviolet) absorption;
   T > 1000 K but gas is rarefied

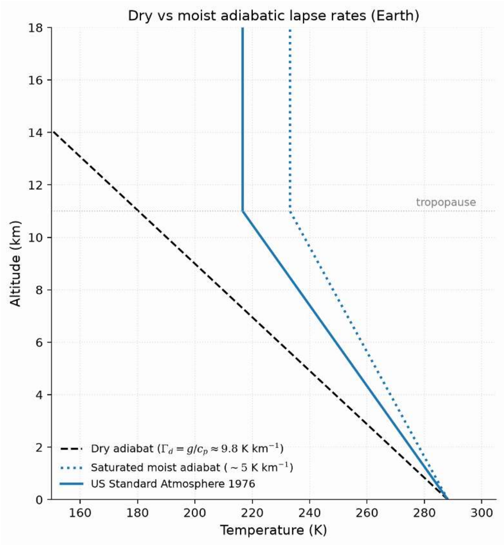
The adiabatic lapse rate

In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

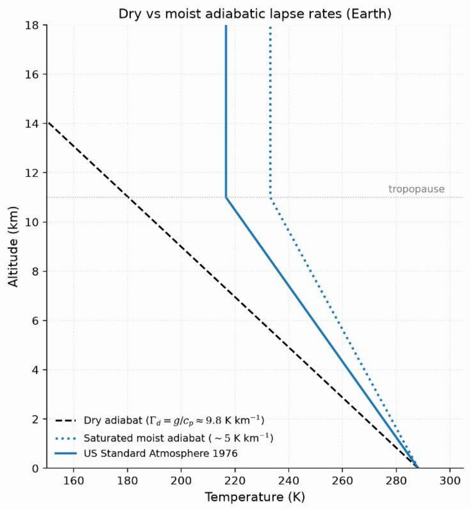
- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8\ \mathrm{K\,km^{-1}}$
- Observed average: ~6.5 K km^{-1} (lower due to latent heat from water condensation)
- Physical picture: rising air expands, does work on surroundings, cools

Tropopause height: ~8 km (poles) to ~17 km (equator).



Dry and moist adiabats against the US Standard Atmosphere 1976 reference profile. Course figure.

Reading the lapse-rate plot



- Dashed: the dry adiabat $\Gamma_d = g/c_p \approx 9.8$ K km^{-1}; dotted: a representative moist adiabat near 5 K km^{-1}.
- Solid: the US Standard Atmosphere 1976 reference profile; its 6.5 K km^{-1} tropospheric lapse rate lies between the two limits because real convection is partly saturated.
- Above the tropopause (~11 km) the adiabats no longer apply: the profile turns nearly isothermal.

Latent heat makes the moist adiabat shallower; the real troposphere sits between the dry and moist limits.

The moist adiabatic lapse rate

When a parcel rises and condensation occurs, latent heat release warms it relative to the dry adiabat.

$$\Gamma_{\mathrm{moist}} = \Gamma_{\mathrm{dry}} \cdot \frac{1 + \dfrac{L_v\, r_s}{R_d\, T}}{1 + \dfrac{L_v^2\, r_s}{c_p\, R_v\, T^2}}$$

- r_s: saturation mixing ratio; L_v: latent heat of vaporisation
- Earth's warm lower troposphere: $\Gamma_{\mathrm{moist}} \approx 5$ K/km; the observed mean (6.5) lies between moist and dry
- Moist convection is **deeper and more vigorous** than dry: latent heat supplies the buoyancy that lifts tropical convection to the tropopause, and drives thunderstorms and hurricanes

Source: Pierrehumbert (2010); Catling & Kasting (2017)

Comparative vertical structures

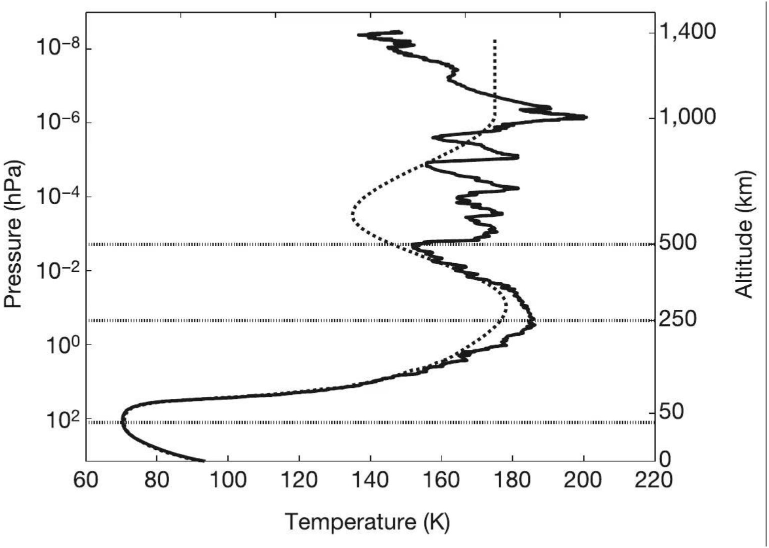
Venus: Massive troposphere (~65 km). No ozone layer → no stratospheric inversion. Cloud deck at 48–70 km.

Mars: Thin troposphere (~40 km), directly into thermosphere. No significant ozone. Dust dominates atmospheric heating.

Jupiter: Troposphere extends hundreds of km deep. Stratosphere heated by CH_4. No solid surface; pressure increases continuously.

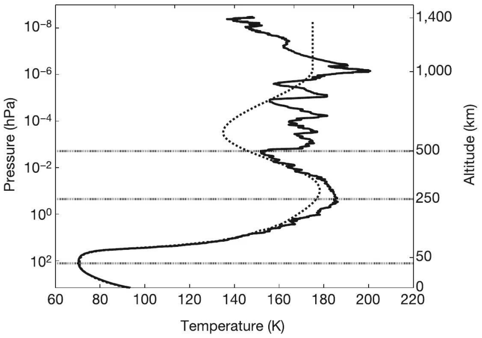
Titan: Thick troposphere (~40 km). Stratosphere heated by organic haze. Extended thermosphere reaches ~1400 km (weak gravity).

Detached haze layers on Titan’s limb, Cassini ISS (PIA06160). Credit: NASA/JPL/Space Science Institute.



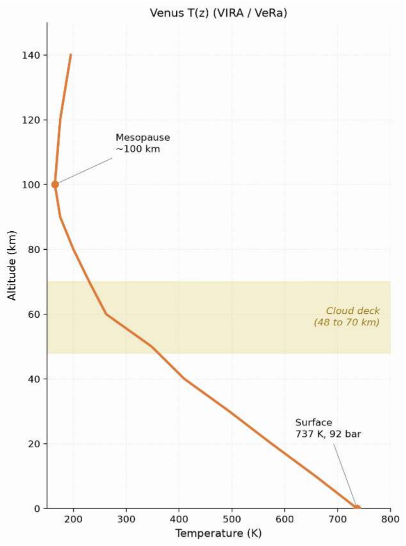
Titan's temperature profile from the Huygens HASI descent. Fulchignoni et al. (2005).

Reading Titan’s descent profile



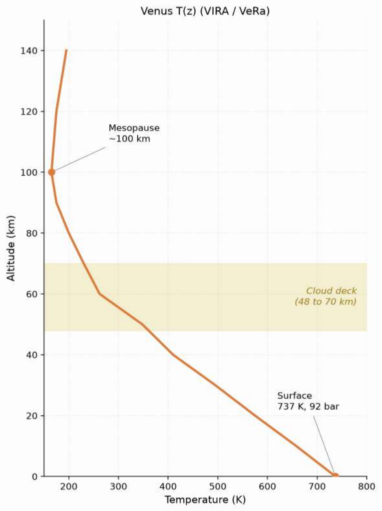
- One instrumented descent (14 January 2005) measured $T(z)$ from 1400 km to the surface.
- The markers fix the layers: tropopause 70.43 K at 44 km, stratopause 186 K at 250 km, mesopause 152 K at 490 km.
- The wiggles above 250 km are gravity waves (buoyancy oscillations, not gravitational waves).

A single descent profile fixed Titan’s entire thermal structure, layer by layer.



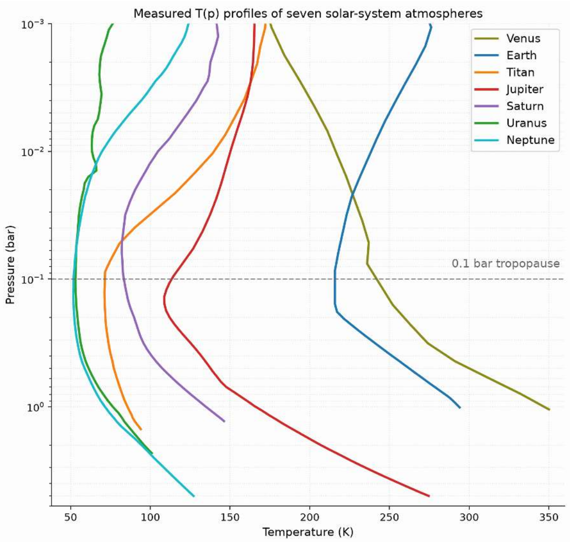
Venus temperature profile, Pioneer Venus/VIRA and Venus Express. Course figure.

Reading Venus's profile



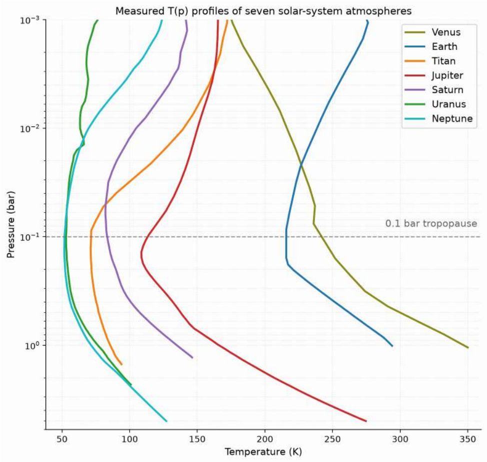
- The profile falls monotonically from the 737 K, 92 bar surface to the mesopause near 100 km: no stratospheric inversion, because Venus has no ozone layer.
- The cloud deck (48 to 70 km, shaded) is where the albedo of 0.77 is made.
- The deep troposphere is near-adiabatic: efficient convection in an optically thick $\mathrm{CO_2}$ atmosphere.

No ozone, no inversion: the temperature falls monotonically from the surface to the mesopause.



Measured temperature-pressure profiles of seven solar-system atmospheres. Digitized from Robinson & Catling (2014), Fig. 1.

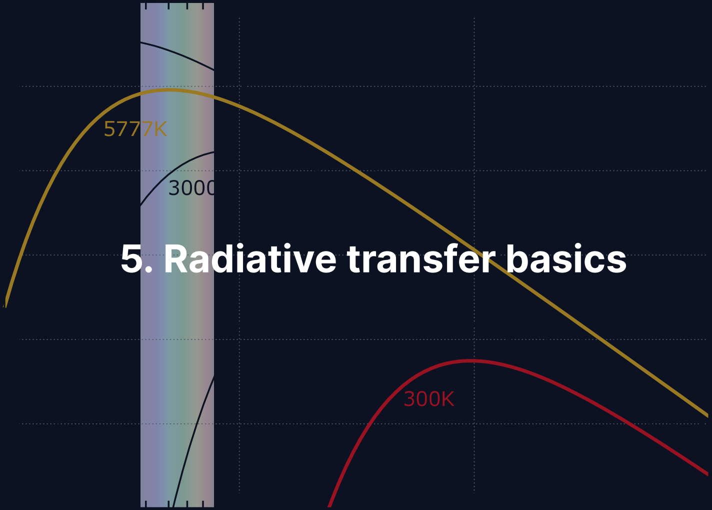
Reading the 0.1-bar rule



- Seven atmospheres, one pressure axis: the tropopause minimum sits near 0.1 bar on Earth, Titan, and all four giants; Venus lacks a strong stratospheric inversion and shows only a weak minimum.
- Deeper than the 0.1 bar level, the atmosphere is optically thick to thermal IR and convects; higher up, it turns transparent and radiative.
- Any UV absorber aloft then builds a stratospheric inversion on top of the same minimum.

Near 0.1 bar, thick atmospheres switch from convective to radiative: a near-universal tropopause.

5. Radiative transfer basics



Blackbody spectra. Credit: Wikimedia, public domain

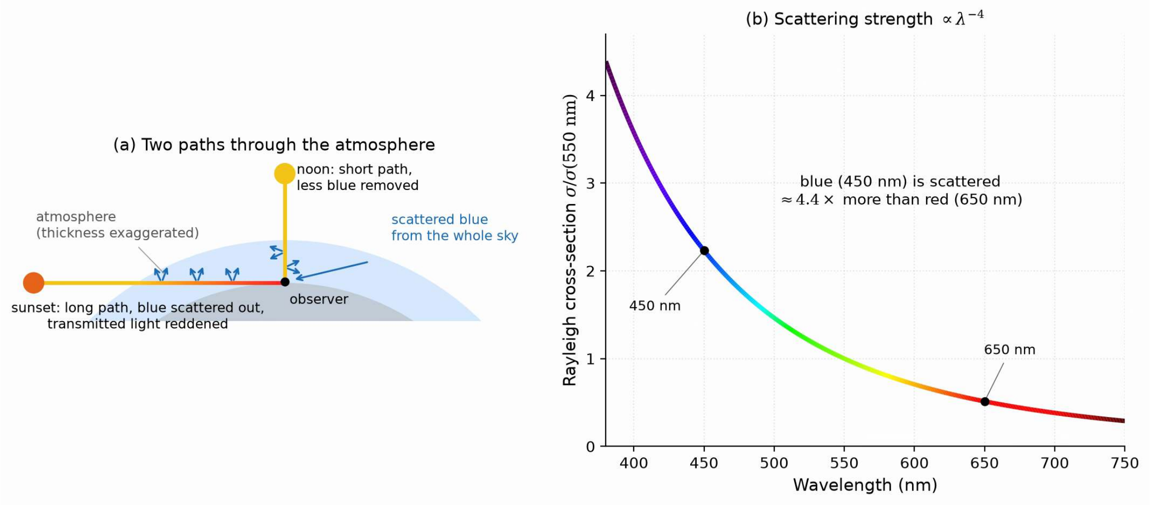
Three interactions: radiation and matter

When radiation passes through an atmosphere:

Absorption: Photon energy → internal energy (vibrational, rotational)
Key absorbers: $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{CH_4}$, $\mathrm{O_3}$, $\mathrm{N_2O}$

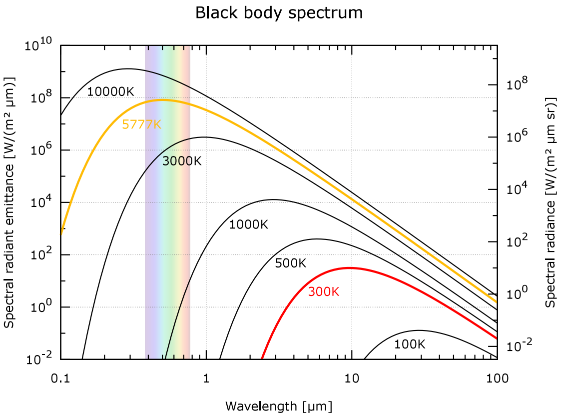
Emission: Warm gas radiates at wavelengths where it absorbs (Kirchhoff's law)
This is how the atmosphere radiates energy to space

Scattering: Photons redirected, not absorbed
Rayleigh ($\propto \lambda^{-4}$): blue sky, red sunsets
Mie (larger particles): less wavelength-dependent



Rayleigh scattering: the long sunset path removes blue light from the beam, and the scattering strength scales as λ^{-4}. Course figure.

Stellar vs. planetary emission



Credit: Wikimedia, public domain

- Sun (~5800 K): peak at ~0.5 μm (visible)
- Planet (~300 K): peak at ~10 μm (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.

Optical depth

The cumulative absorption along a path through the atmosphere:

$$\tau = \int_0^s \kappa\, \rho\, \mathrm{d}s'$$

- κ = mass absorption coefficient ($\mathrm{m^2\,kg^{-1}}$)
- τ is dimensionless: counts "e-folding lengths" of absorption

Optically thin ($\tau \ll 1$):
Most radiation passes through.

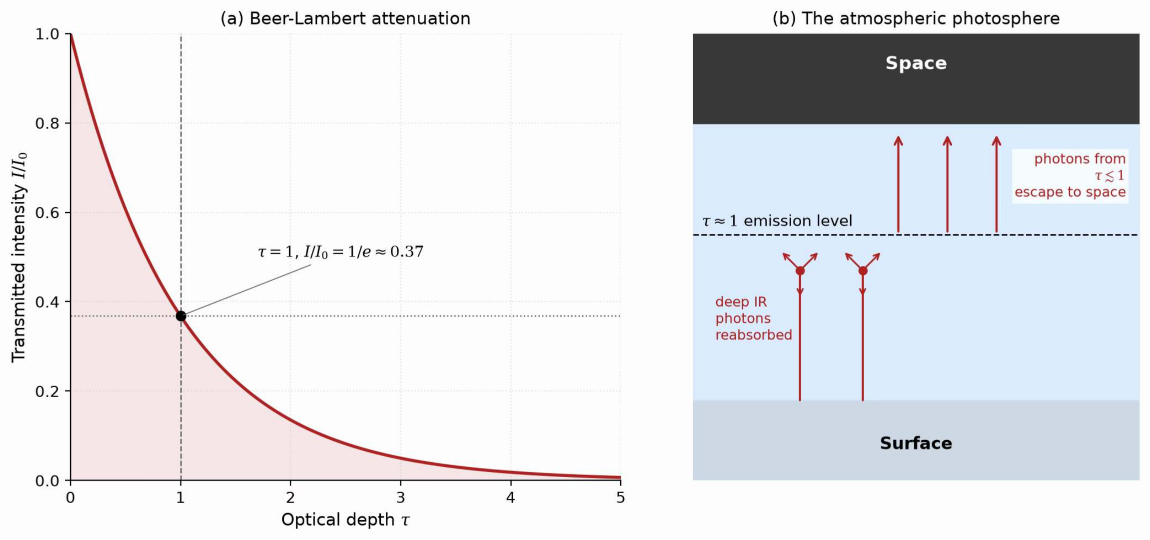
Optically thick ($\tau \gg 1$):
Radiation strongly absorbed; only photons from near the "surface" escape.

The Beer–Lambert law

For a beam of initial intensity I_0 through a medium of optical depth τ:

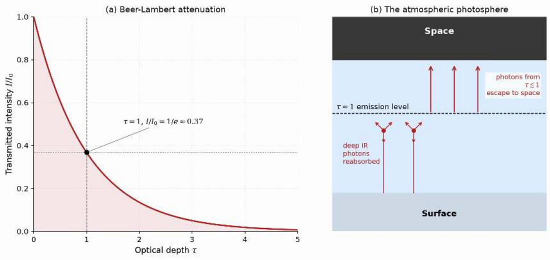
$$\boxed{I = I_0 \, e^{-\tau}}$$

- Intensity drops by factor e per unit optical depth
- **Atmospheric photosphere:** at $\tau \approx 1$, photons escape to space
- Below $\tau = 1$: photons reabsorbed before escaping
- Above $\tau = 1$: photons escape freely
- Temperature at the $\tau \approx 1$ level sets T_{eff}



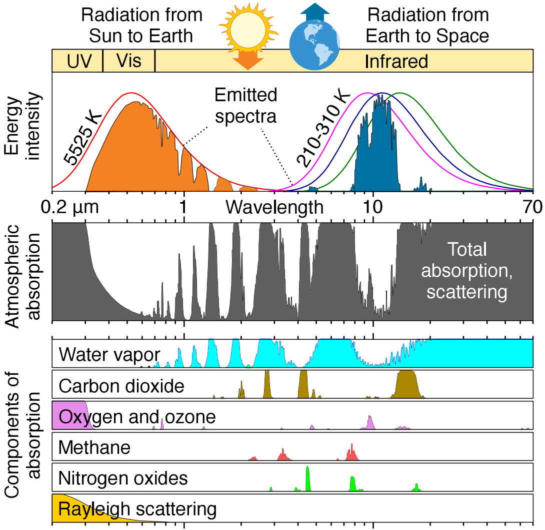
The atmospheric photosphere: Beer-Lambert attenuation and the $\tau \approx 1$ emission level. Course figure.

Reading the photosphere schematic



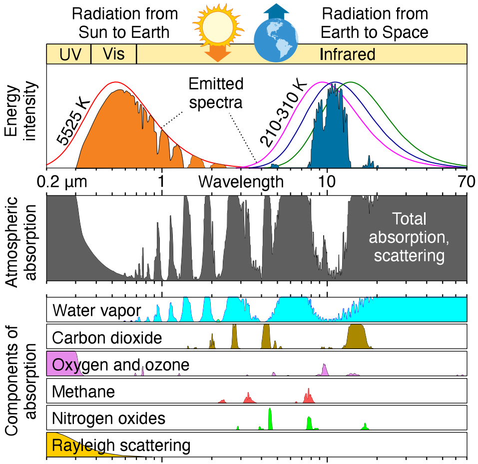
- Panel (a): transmitted intensity falls as $e^{-\tau}$; the $\tau = 1$ level transmits $1/e \approx 0.37$.
- Panel (b): IR photons emitted deep down ($\tau \gg 1$) are reabsorbed; only photons from $\tau \lesssim 1$ reach space.
- The temperature at the $\tau \approx 1$ level is what an observer from space measures: the planet's effective temperature.

A planet, like a star, has a photosphere: the $\tau \approx 1$ surface from which its photons escape.



Earth's atmospheric absorption spectrum from UV to IR. Credit: Wikimedia Commons, public domain.

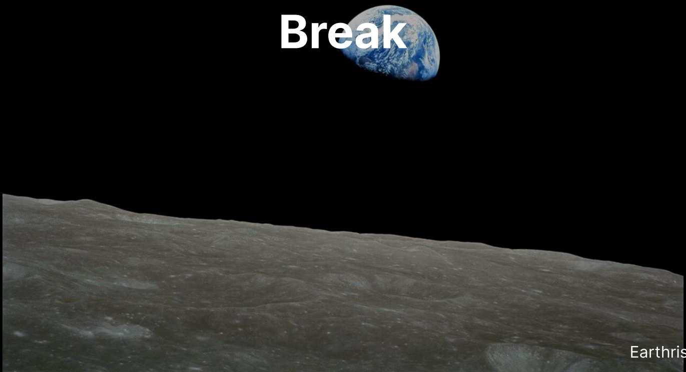
Reading the absorption spectrum



- Top: the incoming solar spectrum; below: the absorption bands of each gas separately.
- H_2O and CO_2 blanket the thermal infrared, right where a ~288 K surface emits.
- The 8–13 μm atmospheric window stays relatively transparent: it is where Earth radiates most efficiently to space.

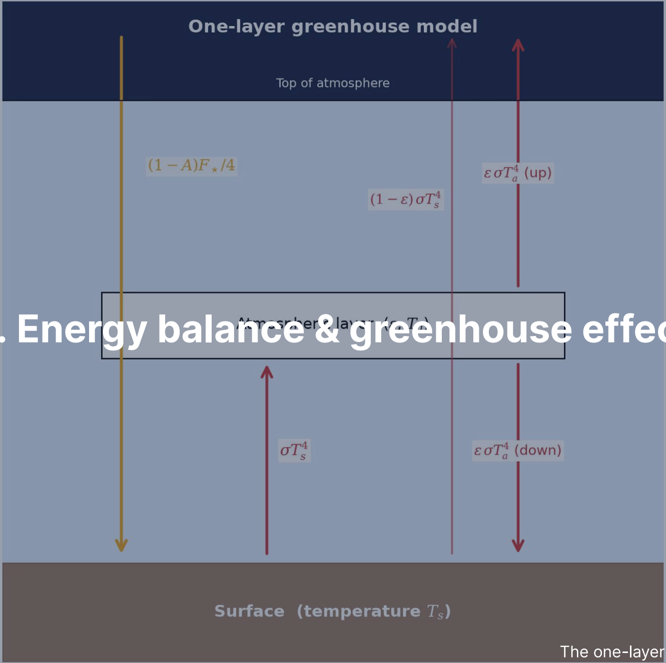
Transparent to sunlight, opaque to surface IR: the asymmetry that powers the greenhouse effect.

Break



Earthrise, Apollo 8 (1968). Credit: NASA

6. Energy balance & greenhouse effect



The one-layer greenhouse model. Course figure

Radiative-convective equilibrium

A realistic atmosphere combines two regimes:

- **Troposphere:** strong greenhouse absorption → unstable to convection
  - Temperature follows the adiabatic lapse rate
  - Most of the atmospheric mass
  - Clouds and weather occur here
- **Stratosphere:** optically thin → radiative equilibrium
  - Temperature set by the balance of absorbed vs. emitted radiation
  - Stable against convection
  - On Earth, warmed by ozone UV absorption

The boundary (the tropopause) occurs where the radiative gradient becomes less steep than the adiabat.

Manabe and Strickler (1964)

The first realistic 1D radiative-convective model of Earth's atmosphere.

- Solved for the vertical temperature profile with absorbing gases (H_2O, CO_2, O_3)
- Reproduced the surface temperature and the tropopause self-consistently
- Extended in 1967 to a fixed distribution of relative humidity, giving the first estimate of the warming from doubled CO_2
- Foundation of modern climate modelling
- Manabe shared the 2021 Nobel Prize in Physics for the physical modelling of Earth's climate

Source: Manabe & Strickler (1964); Manabe & Wetherald (1967), *J. Atmos. Sci.*

Planetary energy balance

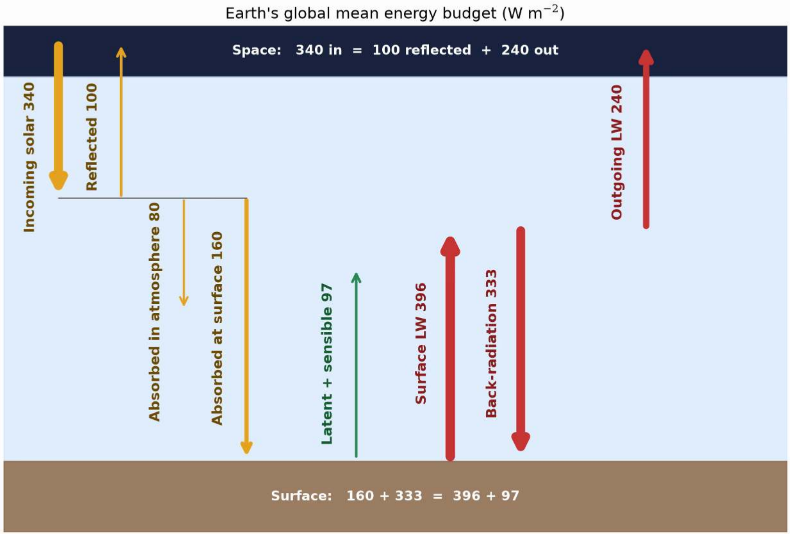
Absorbed stellar power = emitted thermal radiation:

$$(1 - A)\frac{L_\star}{4\pi d^2}\,\pi R_p^2 = 4\pi R_p^2\,\sigma\,T_{\mathrm{eff}}^4$$

- A = Bond albedo (fraction of light reflected)
- πR_p^2 = cross-sectional area (intercepts light)
- $4\pi R_p^2$ = total surface area (emits isotropically)

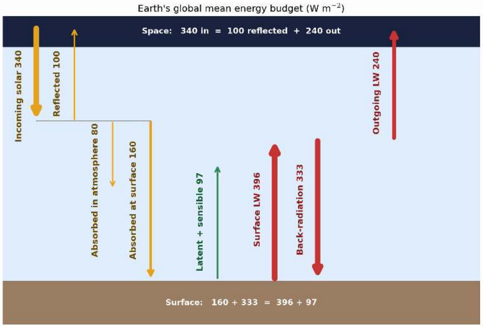
Solving:

$$\boxed{T_{\mathrm{eff}} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\,d^2}\right]^{1/4}}$$



Earth's globally averaged energy budget in W m^{-2}. After Trenberth et al. (2009).

Reading the energy budget



- Of 340 $W\,m^{-2}$ of sunlight, 100 is reflected (A = 0.30), 80 is absorbed in the atmosphere, 160 at the surface.
- The surface emits 396 $W\,m^{-2}$ in the infrared; the atmosphere returns 333 $W\,m^{-2}$ as back-radiation.
- Latent and sensible heat (97 $W\,m^{-2}$) close the surface budget; at the top, 240 $W\,m^{-2}$ of IR leaves.

Back-radiation, not direct sunlight, is the largest flux the surface receives: 333 $W\,m^{-2}$ of recycled solar energy versus 160 direct.

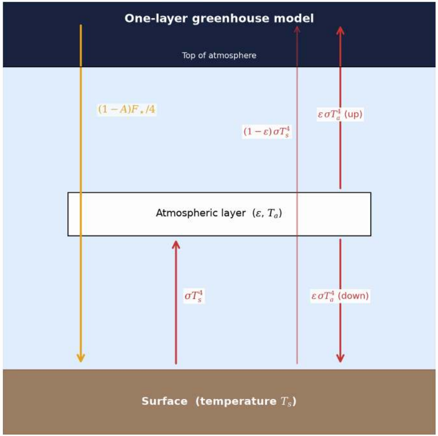
Effective vs. surface temperature

Body	A	d **(AU)**	T_{eff} **(K)**	T_{surf} **(K)**	ΔT **(K)**
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\mathrm{surf}} - T_{\mathrm{eff}}$ reveals the **greenhouse effect**.
Venus: 510 K. Earth: 33 K. Mars: ~5 K (only 6 mbar, no water-vapour amplifier).

The greenhouse mechanism



Credit: Course figure

1. Sunlight (visible, ~0.5 μm) reaches the surface
2. Surface emits thermal IR (~10–15 μm)
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

Result: $T_{\mathrm{surf}} > T_{\mathrm{eff}}$

One-layer greenhouse model: setup

Single isothermal layer, emissivity ε in IR, transparent at visible:

Atmospheric balance:

$$\varepsilon\,\sigma\,T_s^4 = 2\,\varepsilon\,\sigma\,T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{T_s^4}{2}$$

Surface balance:

$$(1 - A)\frac{F_\star}{4} + \varepsilon\,\sigma\,T_a^4 = \sigma\,T_s^4$$

Top-of-atmosphere:

$$(1 - A)\frac{F_\star}{4} = \sigma\,T_{\mathrm{eff}}^4$$

One-layer greenhouse model: result

Combining the three balance equations:

$$\boxed{T_s = T_{\mathrm{eff}}\left(\frac{2}{2-\varepsilon}\right)^{1/4}}$$

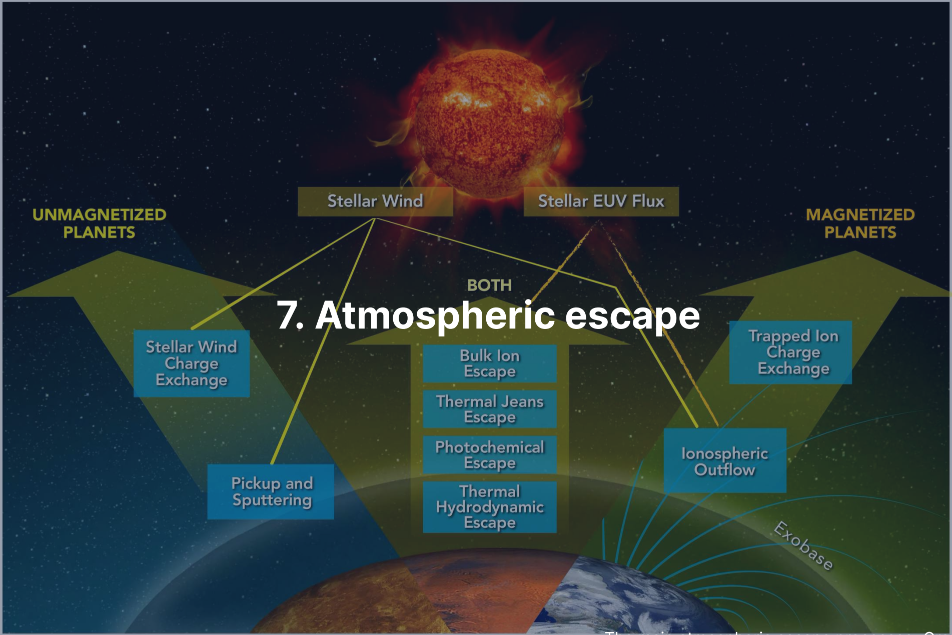
$\varepsilon = 0$ (no greenhouse): $T_s = T_{\mathrm{eff}}$

$\varepsilon = 1$ (perfect absorber): $T_s = 2^{1/4}\,T_{\mathrm{eff}} \approx 1.19\,T_{\mathrm{eff}}$

For Earth: $T_s \approx 1.19 \times 255 \approx 303$ K. Reasonable first estimate.

Looking further ahead: What happens when T_s rises so high that outgoing radiation saturates? → *Runaway greenhouse* (Lecture 9).

7. Atmospheric escape



The main atmospheric escape processes. Gronoff et al. (2020)

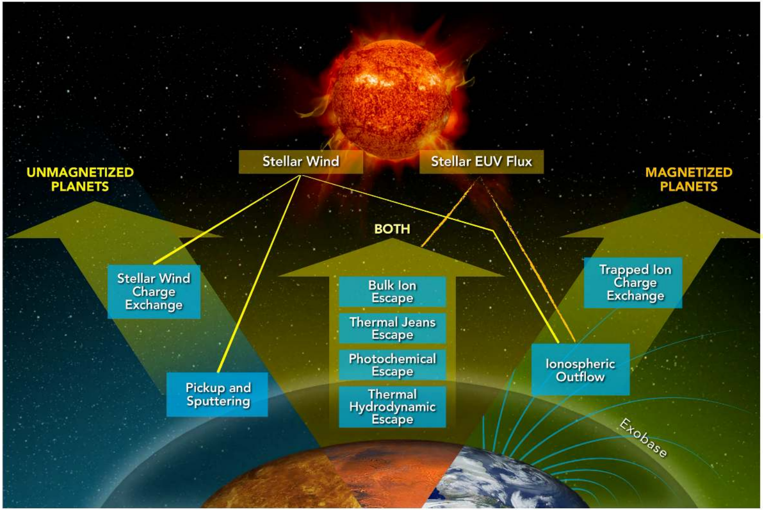
Thermal (Jeans) escape

Molecules have a Maxwell–Boltzmann velocity distribution. The high-velocity tail extends beyond escape speed.

The **Jeans escape parameter** at the exobase (the level where collisions become rare; defined later in this section):

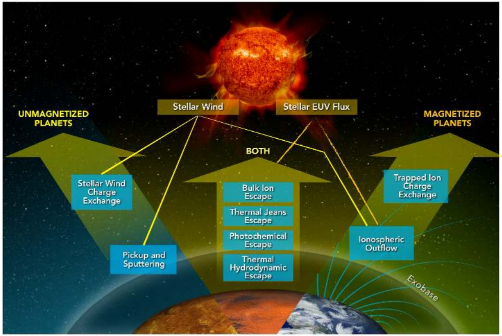
$$\lambda_J = \frac{GM_p m}{k_\mathrm{B} T_\mathrm{exo} r_\mathrm{exo}} = \frac{v_\mathrm{esc}^2}{v_\mathrm{th}^2}$$

- $\lambda_J \gg 1$: escape negligible (tail too depleted)
- $\lambda_J \lesssim 2$–3: rapid atmospheric erosion
- Escape flux: $\Phi_J \propto (1 + \lambda_J)\, e^{-\lambda_J}$, exponentially sensitive



The main atmospheric escape processes. Gronoff et al. (2020).

Reading the escape processes



- Centre: thermal (Jeans, hydrodynamic), photochemical, and bulk ion escape act on every planet, powered mainly by stellar EUV.
- Left, unmagnetised (Venus, Mars): the stellar wind hits the atmosphere directly; sputtering, ion pickup, charge exchange.
- Right, magnetised (Earth): the field deflects the wind but opens ionospheric outflow and trapped-ion charge exchange.

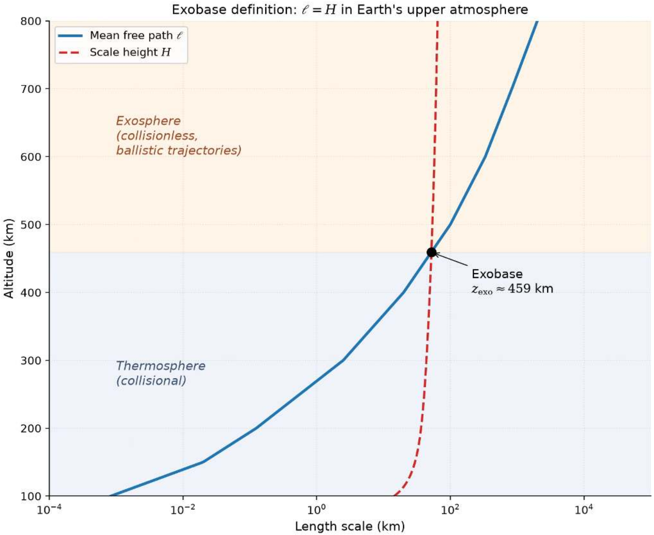
A magnetic field does not switch escape off; it selects which loss processes operate.

Jeans parameter for Earth and Mars

Exobase temperatures: 1000 K (Earth), 270 K (Mars). Earth's varies from ~600 K at solar minimum to ~1500 K at solar maximum.

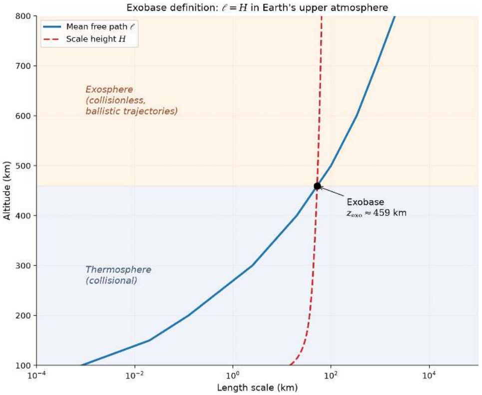
Species	m (u)	λ_J (Earth)	λ_J (Mars)
H	1	7.0	5.3
H_2	2	14	11
He	4	28	21
N_2	28	200	150
CO_2	44	310	230

Heavy species (N_2, CO_2): λ_J so large that Jeans escape is negligible.
Atomic H: moderate λ_J → both Earth and Mars lose hydrogen to space.



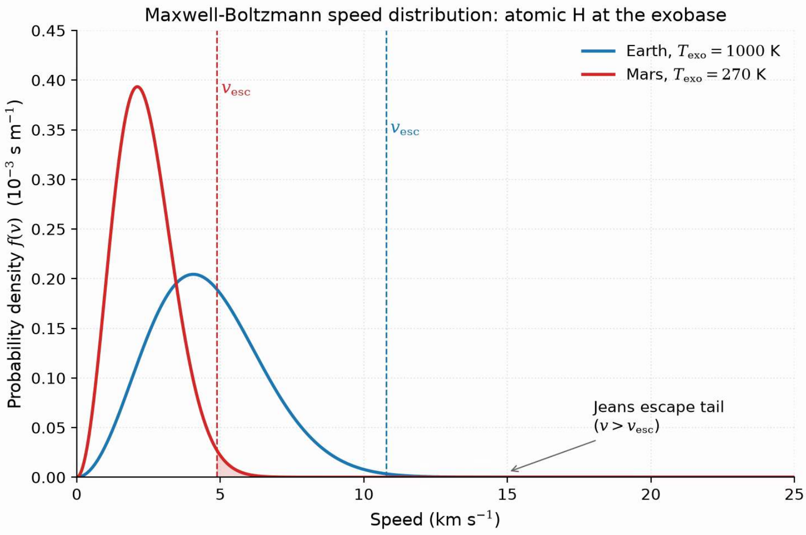
The exobase: where the mean free path (distance between molecular collisions) overtakes the scale height. Course figure.

Reading the exobase crossing



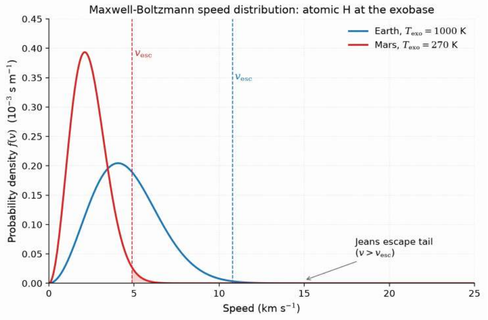
- The mean free path $\ell = 1/(n\sigma)$ climbs exponentially as density falls; the scale height H barely changes.
- They cross near 459 km in this Earth profile: below, collisions make a fluid; above, molecules fly ballistically. On Mars it is near 200 km.
- A molecule crossing upward faster than $v_{\rm esc}$ never collides again: it is gone.

One crossing splits the atmosphere into fluid below and free flight above; escape happens at that boundary.



Maxwell-Boltzmann speed distributions for atomic H at the Earth and Mars exobases. Course figure.

Reading the speed distributions



- Atomic hydrogen at two temperatures: Earth's hot exobase (1000 K, blue) and Mars's cold one (270 K, red).
- The dashed lines mark v_{esc} at each exobase: 10.8 and 4.9 km s^{-1}; only the shaded tails beyond them escape.
- The tail area scales as $e^{-\lambda_J}$: modest changes in temperature or gravity swing the escape flux by orders of magnitude.

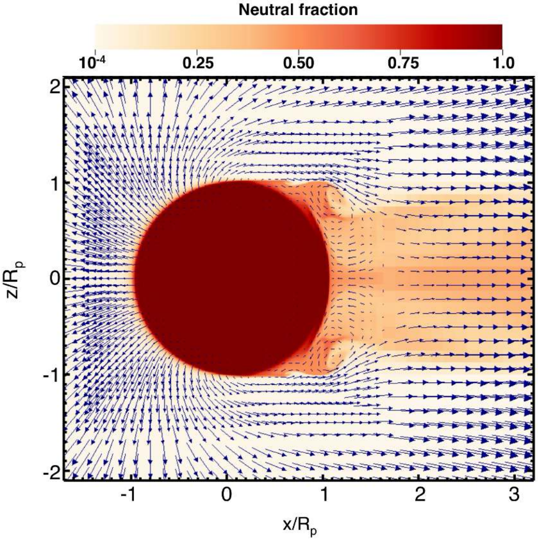
Escape is decided in the exponential tail of the speed distribution: λ_J fixes the fate of any species in the Jeans regime.

Hydrodynamic escape

When stellar EUV heating is intense, escape transitions from molecule-by-molecule to a bulk **atmospheric wind**:

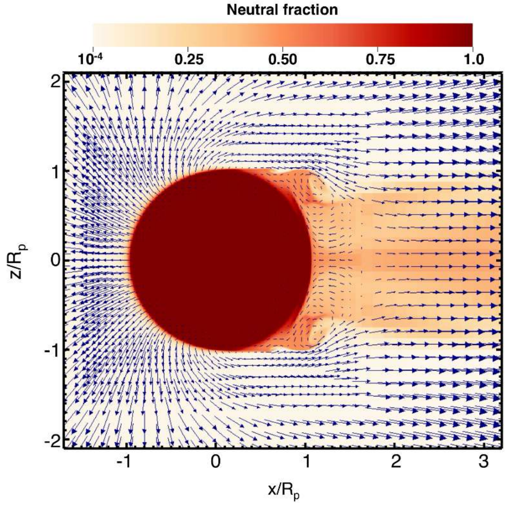
- Most important in the first few hundred Myr (young stars: 10–100× more EUV)
- Can strip H_2-rich primary atmospheres from planets up to several $M_\oplus$
- Outflowing H can **drag along heavier species** (He, C, N, O)
- Leading explanation for the exoplanet **radius valley** (deficit at ~1.5–2 $R_\oplus$; Lecture 13)
- Seen directly: HD 209458 b (H), GJ 436 b (H), WASP-107 b (He)

Source: Vidal-Madjar et al. (2003); Ehrenreich et al. (2015); Spake et al. (2018)



Simulated hydrodynamic outflow from a hot Jupiter. Owen (2019); simulation by Tripathi et al. (2015).

Reading the outflow simulation



- Colour: neutral hydrogen fraction; arrows: flow velocity. The star is to the left; axes in planetary radii.
- Day-side gas is heated, ionised, and accelerated away as a wind that wraps around the terminator.
- A partially neutral wake trails on the night side; the shear against it produces Kelvin-Helmholtz rolls near $x \approx 1\,R_p$.

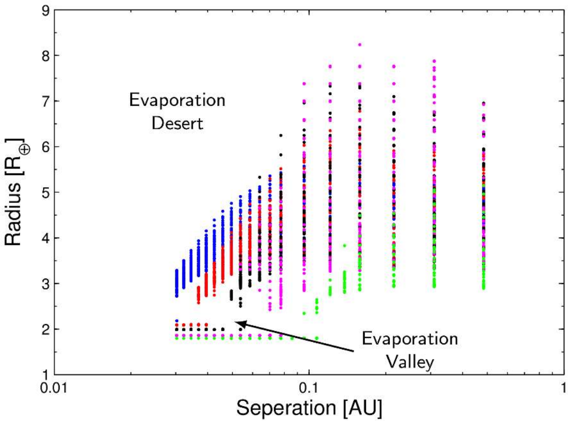
The whole upper atmosphere leaves as a fluid wind, not a molecule-by-molecule leak.

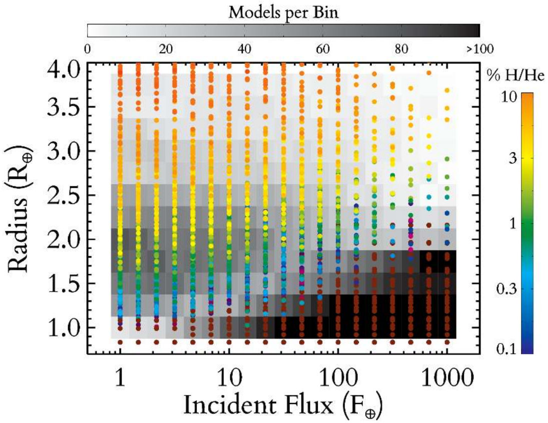
Stripping primary atmospheres

Young stars are highly active in EUV and X-ray; primordial H/He envelopes can evaporate.

- EUV flux is ~100× higher in the first few hundred Myr than today
- Mass loss rate for hydrodynamic regime: $\dot{M} \propto F_{\mathrm{EUV}}\, R_p^3/(GM_p)$
- For sub-Neptunes ($M_p \lesssim 10\, M_\oplus$), atmospheres can be fully stripped in a few hundred Myr
- For Neptune-mass planets and above, envelopes are retained
- Produces the "evaporation valley" at ~1.8 $R_\oplus$ in Kepler data

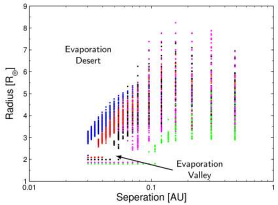
Source: Owen & Wu (2013); Lopez & Fortney (2013); Fulton et al. (2017)

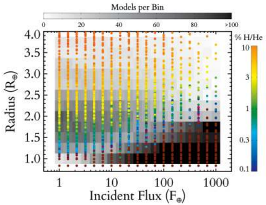




Photoevaporation sculpting the close-in exoplanet population. Left panel from Owen & Wu (2013), right panel from Lopez & Fortney (2013); the desert and valley labels follow Owen (2019).

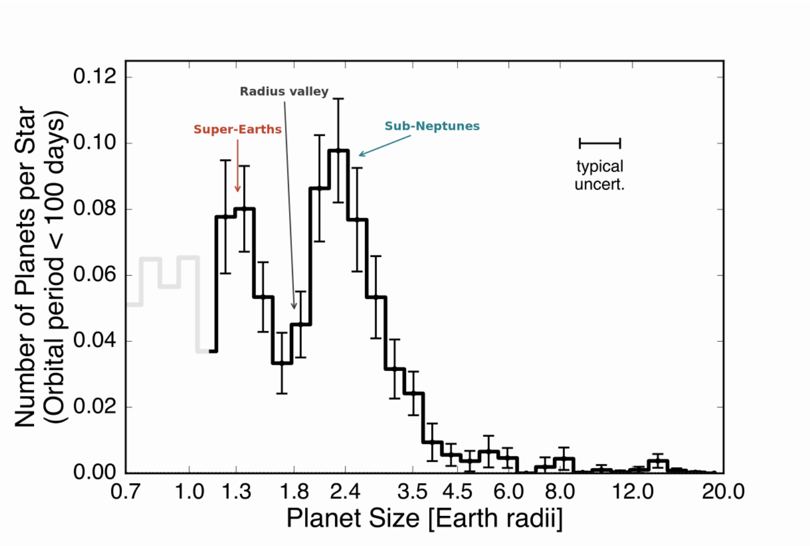
Reading the evaporation model





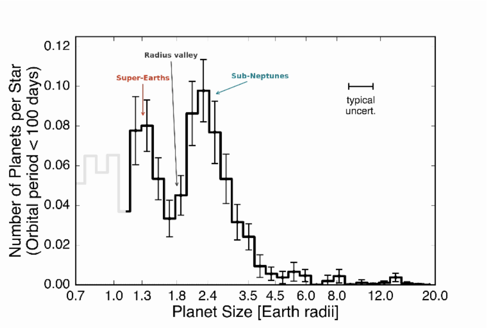
- Left: after 10 Gyr of EUV-driven escape, model planets split into stripped cores and envelope-keepers, leaving an empty valley near 1.5–2 $R_\oplus$.
- The evaporation desert at small separations is fully stripped; close to the star no envelope survives.
- Right: the survivors, coloured by retained H/He mass fraction, a few percent is enough to double a radius.

A few hundred Myr of stellar EUV decides whether a planet stays a sub-Neptune or is stripped to its core.



The observed radius valley in the California-Kepler Survey. After Fulton et al. (2017).

Reading the radius valley



- The distribution is bimodal: super-Earths (rocky cores) near 1.3 $R_\oplus$, sub-Neptunes near 2.4 $R_\oplus$.
- The gap near 1.8 $R_\oplus$ is exactly where photoevaporation models predict the valley; the grey part of the histogram, below about 1.1 $R_\oplus$, marks low completeness, not a physical feature.
- The two peaks are the two endpoints: stripped rocky cores and cores that kept a few percent of H/He.

The data go missing just where escape physics predicted a gap: photoevaporation observed at population scale.

Energy-limited escape rate

A convenient estimate of mass loss:

$$\dot{M}_{\mathrm{EL}} = \eta \frac{\pi F_{\mathrm{EUV}} R_p^3}{G M_p}$$

- F_{EUV}: incident EUV flux
- η: heating efficiency, typically 0.1–0.2
- Scales as R_p^3: large, fluffy atmospheres lose mass fastest
- Scales inversely with M_p: low gravity eases escape
- Valid when recombination is negligible (“energy-limited regime”)

Source: Watson et al. (1981); Owen (2019)

Non-thermal escape mechanisms

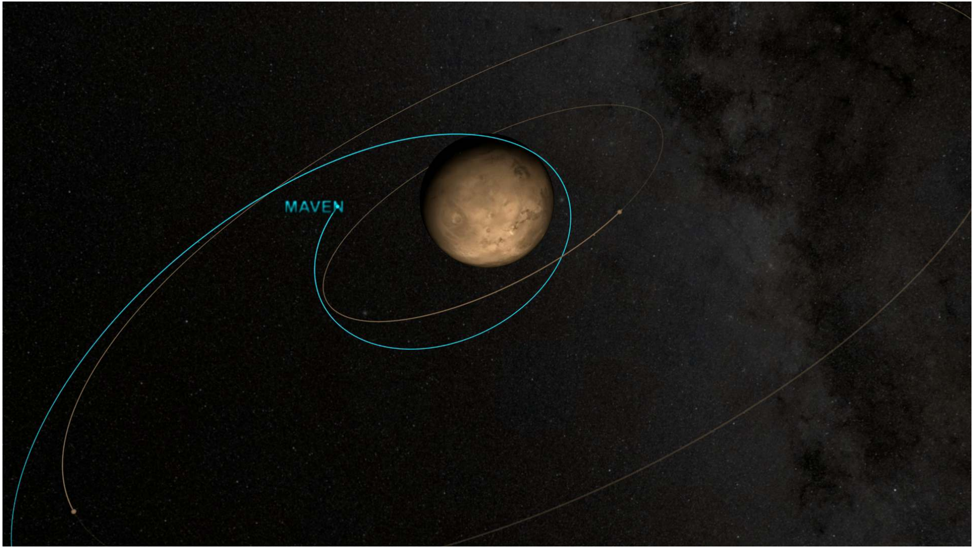
Sputtering: Solar wind ions impact upper atmosphere, eject molecules.
Significant for Mars (no global magnetic field).

Photochemical: UV dissociates $H_2O \rightarrow$ H + OH; fast H escapes.
Key for Venus and Mars water loss.

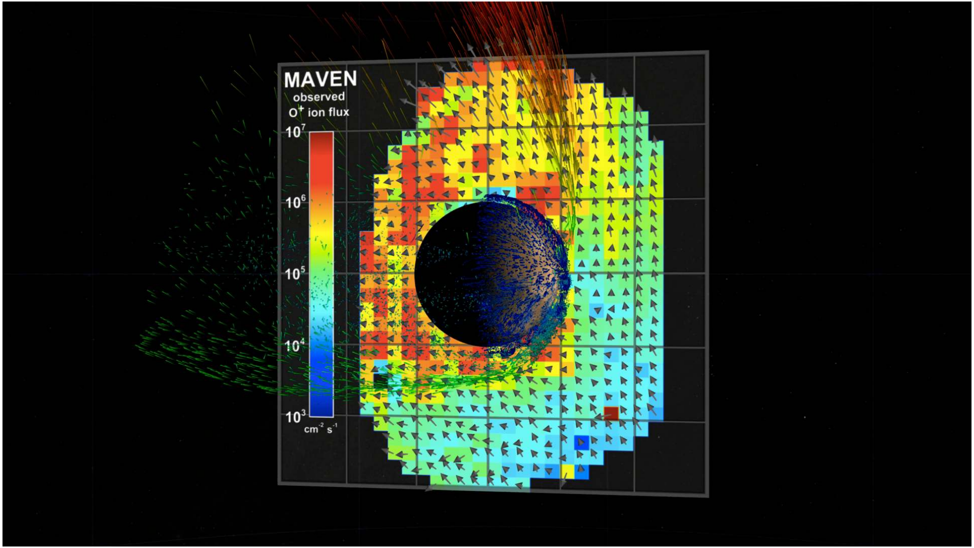
Ion pickup: Atmospheric atoms ionised by UV, swept away by solar wind *B*-field.
Effective at unmagnetised planets.

Impact erosion: Large impacts can eject substantial atmospheric mass in a single event.

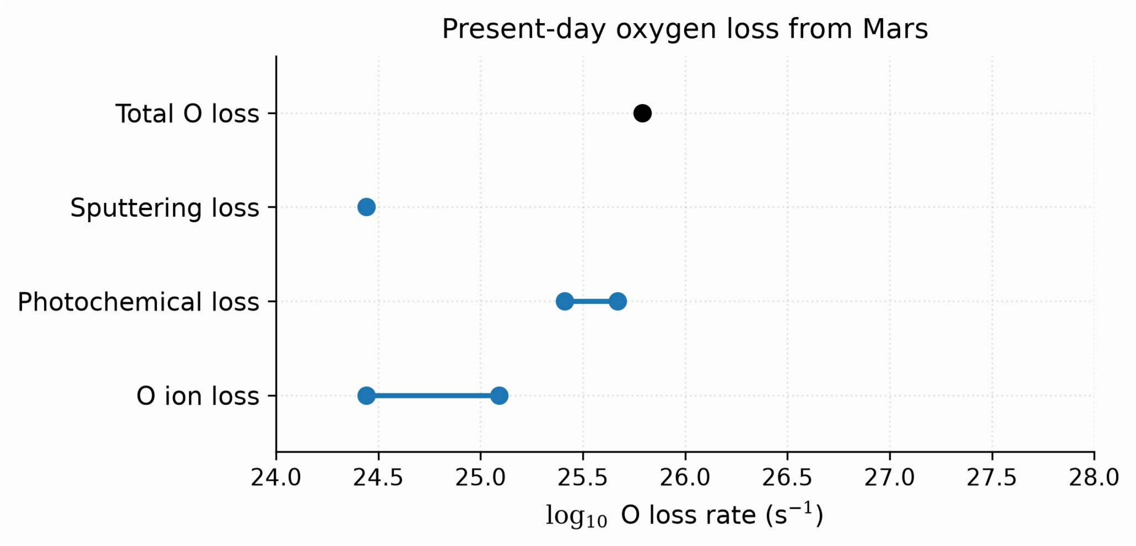
MAVEN at Mars: the O^+ pickup channel alone carries ~100 g s^{-1}; photochemical escape dominates the oxygen budget today.



MAVEN's elliptical science orbit at Mars: periapsis in the thermosphere, apoapsis in the ion tail. Credit: NASA SVS.

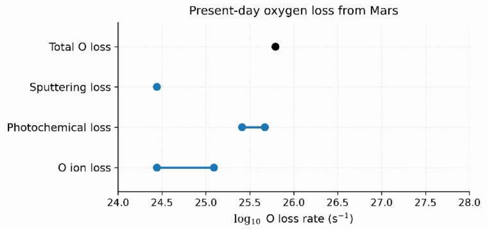


The ion plume escaping Mars, rendered from MAVEN data. Credit: NASA SVS.



Present-day oxygen loss from Mars by escape channel. After Jakosky et al. (2018).

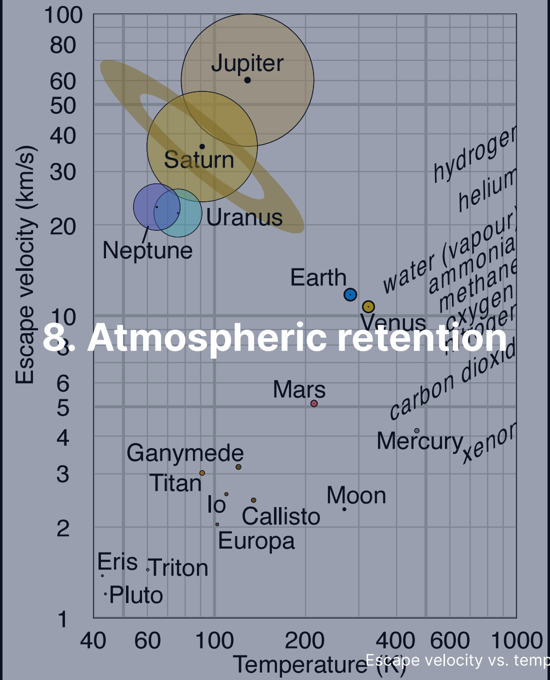
Reading the oxygen loss channels



- Each bar is one escape channel for oxygen: photochemical escape dominates today; ion escape and sputtering are sub-dominant.
- Under the young, EUV-active Sun, sputtering was likely comparable or larger: the channels shift over time.
- Summed with hydrogen escape, the total reaches the 2–3 $kg\,s^{-1}$ MAVEN measures.

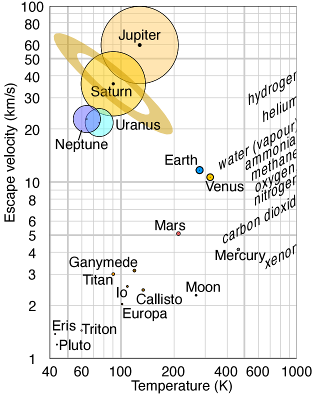
Mars leaks 2 to 3 kilograms of atmosphere every second, and the books balance channel by channel.

8. Atmospheric retention



Escape velocity vs. temperature. Credit: Wikimedia, CC BY-SA 4.0

Atmospheric retention: v_{esc} vs. T diagram



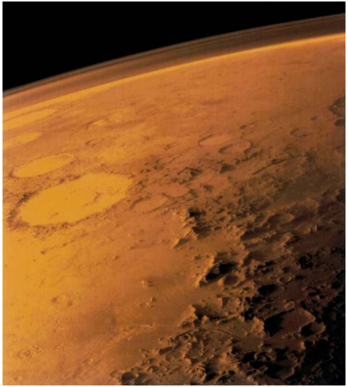
Credit: Wikimedia, CC BY-SA 4.0

Rule of thumb for long-term retention:

$$v_{\mathrm{esc}} \gtrsim 6\, v_{\mathrm{th}}$$

- Gas giants: retain **everything**
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold → retains $\mathrm{N_2}$ despite low v_{esc}
- Moon/Mercury: retain nothing (Mercury: ~10^{-15} bar exosphere)

Mars: a lost atmosphere



Credit: NASA/JPL (Viking 1, 1976)

- Present surface pressure: only 6 mbar
- Too thin for liquid water or significant greenhouse warming
- Geological evidence: ancient rivers, lakes, possibly ocean
- Mars had a much thicker atmosphere >4 Gyr ago
- Dynamo loss (~4.1 Ga, Lecture 4) → unshielded solar wind erosion

9. Recent advances & outlook

Earth’s atmosphere. Credit: NASA/ISS

Recent advances

- **JWST / TRAPPIST-1 b:** Thermal emission consistent with bare rock, no significant atmosphere. Inner rocky planets around active M dwarfs may be stripped.
- **JWST / TRAPPIST-1 c:** Similar result, reinforcing theoretical predictions of enhanced atmospheric escape around low-mass stars.
- **MAVEN at Mars:** Quantified loss rates for multiple species. Ion escape (solar wind stripping) dominates over Jeans escape. Integrated losses can account for a substantial fraction of Mars's early atmosphere.

Central theme: a planet's atmospheric inventory is set by the balance between escape and resupply. Outgassing draws on interior heat and wanes as the interior cools (Lecture 3), so retention depends on the interior thermal evolution, not only on the present-day escape rate.

Impact erosion of atmospheres

Impacts erode atmospheres in two regimes (Lecture 4).

- **Local:** an ordinary impactor expels at most the air above the tangent plane at the impact point, a fraction ~ $H/2R$ of the whole atmosphere
- Erosion is therefore cumulative; km-scale planetesimals are the most efficient eroders per unit mass
- **Global:** a giant impact shakes the whole planet, but even the Moon-forming impact removes only ~20%
- Key process while the bombardment lasts, the first ~100 Myr; EUV-driven escape runs on a slower clock and continues for a few hundred Myr
- Signature: noble gas isotope ratios record impact-driven loss

Source: Schlichting & Mukhopadhyay (2018); Genda & Abe (2003, 2005)

Outgassing vs. escape: a dynamic balance

An atmosphere is not a static inheritance; it is a reservoir in flow.

- **Supply:** volcanic outgassing, impact delivery, biological fluxes
- **Loss:** thermal escape, non-thermal escape, chemical weathering, sequestration in oceans/crust
- The **steady state** depends on the balance of these fluxes
- Earth's active volcanism continually replaces CO_2 on the carbonate-silicate timescale (weathering-outgassing CO_2 cycle; ~0.5 Myr)
- Mars lost its active volcanism and dynamo, so escape was not replenished
- Venus has active (perhaps episodic) volcanism, but no oceans to absorb CO_2

Atmospheric longevity is as much about **replacement** as about initial inventory.

MAVEN at Mars: quantifying escape

NASA's MAVEN (2014–) has measured Mars' atmospheric loss in situ.

- Present-day total escape rate: ~2–3 kg/s (H and O combined)
- Dominant loss channel: photochemical escape of oxygen
- Enhanced during solar storms (~10× increase)
- Integrated loss over 4 Gyr: consistent with 0.5–0.8 bar of CO_2 lost
- Combined with hydrogen escape: up to ~23 m global equivalent layer of water lost
- Young-Sun EUV may also have driven thermal escape of carbon, which blocks a dense early CO_2 atmosphere

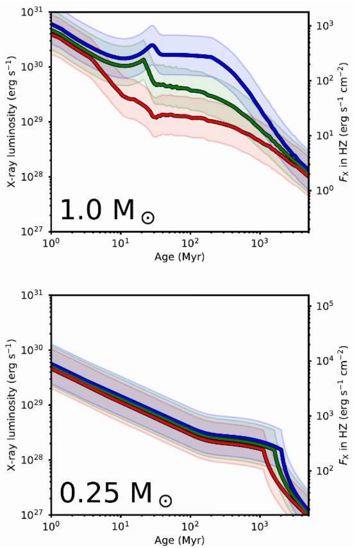
Source: Jakosky et al. (2018), *Icarus*; Lillis et al. (2017); Tian et al. (2009)

JWST and the fate of rocky atmospheres

JWST has begun to directly test whether rocky planets around M dwarfs retain atmospheres.

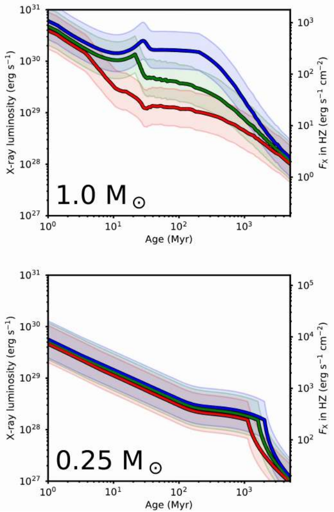
- TRAPPIST-1 b: MIRI thermal emission consistent with a bare rock (dayside 503 K)
- TRAPPIST-1 c: MIRI shows low heat redistribution ⇒ thin or no atmosphere
- LHS 3844 b: similar result using Spitzer data
- Implications: inner rocky planets around active M dwarfs lose their atmospheres
- Open question: whether **outer** TRAPPIST-1 planets retain atmospheres
- JWST will target TRAPPIST-1 e, f during Cycle 3–4

Source: Greene et al. (2023); Zieba et al. (2023), *Nature*



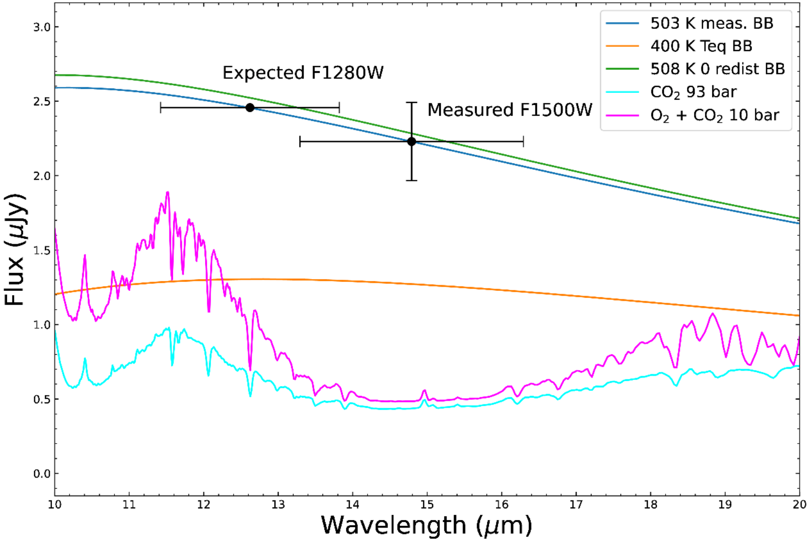
Stellar X-ray luminosity evolution for a Sun-like star and a mid-M dwarf. Johnstone et al. (2021), Fig. 11.

Reading the XUV evolution



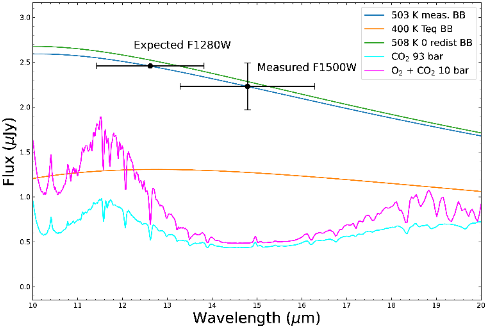
- Two panels, two stellar masses; each curve a rotation history. All stars start saturated at $L_X/L_{\mathrm{bol}} \sim 10^{-3}$.
- A Sun-like star leaves saturation after ~100 Myr; a 0.25 $M_\odot$ M dwarf stays saturated beyond a gigayear.
- Close-in rocky planets around late M dwarfs therefore receive saturated **XUV** (X-ray + EUV) through their entire early history.

Small stars stay active for a gigayear: their planets endure a prolonged escape-driving epoch.



JWST/MIRI 15 μm thermal emission of TRAPPIST-1 b. Greene et al. (2023).

Reading the TRAPPIST-1 b measurement



- The secondary-eclipse depth matches a bare-rock blackbody at 503 K, with no heat redistribution.
- An atmosphere moving heat to the nightside would cool the dayside toward 400 K: too faint.
- Thick $\mathrm{CO_2}$ absorbs at 15 μm and would halve the flux: also excluded.
- Green curve: the 508 K zero-redistribution case. Second black point: the bare-rock fit at 12.8 μm.

A single photometric point rules out a thick atmosphere on TRAPPIST-1 b.

Transmission spectroscopy: preview

During a transit, starlight passes through the planet's **limb**.

- Absorbing molecules imprint dark features in the transmission spectrum
- Path length ~$2\sqrt{2R_pH}$ through a layer of scale height H
- Signal: $\Delta(R_p/R_\star)^2 \sim 2R_pH/R_\star^2$
- Bigger signal for puffy atmospheres (large H) around small stars
- JWST has detected H_2O, CO_2, CH_4, SO_2 in multiple exoplanet atmospheres
- Full treatment in Lecture 13 (Exoplanet Atmospheres)

Source: Seager & Sasselov (2000); Kreidberg et al. (2014)

Runaway greenhouse: a look ahead

What happens when a water-rich atmosphere gets too hot?

- More water vapour → stronger greenhouse → hotter surface
- Positive feedback: hotter surface → more evaporation → more greenhouse
- A water-rich atmosphere cannot radiate more than ~280–310 W/m^2: the **Simpson-Nakajima limit**
- If the absorbed flux exceeds this limit, the oceans evaporate entirely and the runaway becomes unstoppable
- Hydrogen then escapes hydrodynamically, leaving a dry CO_2 atmosphere
- Venus' modern state is the end point of this process

Derived in full in **Lecture 9: Earth & Venus**.

Source: Ingersoll (1969); Nakajima et al. (1992)

Looking ahead to Lecture 6

This lecture treated the atmosphere as a static column: composition, structure, energy balance.

- Lecture 6 (dr. Mara Attia) sets the column in motion
- Condensation and clouds: where the saturation curve is crossed
- Rotation organises flows into circulation cells and jets
- Feedbacks between temperature, ice, and radiation: stable climates and runaway states

> The hydrostatic and radiative results assembled today return in every piece of atmospheric dynamics.

Summary

1. Three atmosphere types: **primary** (captured H_2/He), **secondary** (outgassed), **tertiary** (biologically modified)
2. Hydrostatic equilibrium: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$; leads to exponential pressure decay
3. Scale height: $H = k_\mathrm{B} T / (\mu\, m_u\, g)$; encodes thermal energy vs. gravity
4. Vertical structure: troposphere, stratosphere, mesosphere, thermosphere
5. Beer–Lambert law: $I = I_0\, e^{-\tau}$; $\tau \approx 1$ defines the emission level
6. Energy balance: $T_\mathrm{eff} \propto [(1 - A) L_\star / d^2]^{1/4}$; greenhouse raises T_s above T_eff
7. Escape: Jeans (λ_J), hydrodynamic, non-thermal; retention requires $v_\mathrm{esc} \gtrsim 6\, v_\mathrm{th}$

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`